Homework 5

FIN2020 Fall 2023

Due at Midnight **11:59pm Wednesday Nov. 8, 2023 Beijing Time**

Problem 1

For the following problem please refer to Table 1

Table 1: **Historical Annual Volatilities and Correlations for Selected Stocks (based on monthly returns, 1996-2008).**

	Microsoft	Dell	Alaskan Air	Southwest Airlines	Ford Motor	General Motors	General Mills
Volatility (StDev)	37%	50%	38%	31%	42%	41%	18%
Correlation with:							
Microsoft	1.00	0.62	0.25	0.23	0.26	0.23	0.10
Dell	0.62	1.00	0.19	0.21	0.31	0.28	0.07
Alaska Air	0.25	0.19	1.00	0.30	0.16	0.13	0.11
Southwest Airlines	0.23	0.21	0.30	1.00	0.25	0.22	0.20
Ford Motor	0.26	0.31	0.16	0.25	1.00	0.62	0.07
General Motors	0.23	0.28	0.13	0.22	0.62	1.00	0.02
General Mills	0.10	0.07	0.11	0.20	0.07	0.02	1.00

(a) What is the covariance between the returns for Alaskan Air and General Mills?

(b) What is the volatility of a portfolio with

i. equal amounts invested in these two stocks?

ii. 20% invested in Alaskan Air and 80% invested in General Mills?

iii. 80% invested in Alaskan Air and 20% invested in General Mills?

Problem 2

Consider a portfolio of two stocks. Data shown in Table 2. Let x denote the weight on Stock A and $1 - x$ denote the weight on Stock B. Correlation coefficient equals ρ_{AB}.

The risk-free rate is $r_f = 3\%$.

Table 2:

Stock	Expected Return	Volatility
Stock A	15%	40%
Stock B	7%	30%

(a) What is the minimum variance portfolio when $\rho_{AB} = 0$? What is its expected return and volatility?

(b) What is the minimum variance portfolio when $\rho_{AB} = 0.4$? What is its expected return and volatility?

(c) What is the minimum variance portfolio when $\rho_{AB} = -0.4$? What is its expected return and volatility?

Problem 3

Consider an investor with preferences over the mean and variance of the returns on his or her portfolio that are described by the utility function

$$U(\mu_P, \sigma_P^2) = \mu_P - \left(\frac{A}{2}\right)\sigma_P^2,$$

where a higher value of A corresponds to a larger aversion to risk. Suppose that this investor is able to form a portfolio from a risk-free asset with return r_f and a risky asset with expected return equal to μ_r and a standard deviation of its return equal to σ_r. In this context, the risky asset may simply be the only risky asset available to the investor or it can be the tangency portfolio that represents the optimal combination of many individual risky assets. Either way, our analysis from class showed that the relationship between μ_P and σ_P for the portfolio that combines the risk-free asset with the risky asset is linear:

$$\mu_P = r_f + \left(\frac{\mu_r - r_f}{\sigma_r}\right)\sigma_P.$$

By substituting this constraint into the investor's utility function, his or her problem can be solved as one of choosing σ_P to maximize

$$r_f + \left(\frac{\mu_r - r_f}{\sigma_r}\right)\sigma_P - \left(\frac{A}{2}\right)\sigma_P^2.$$

Use the first-order condition for this problem to solve for the optimal choice of σ_P.

Our analysis from class also showed that the portfolio formed from the risk-free and risky asset that has a return with standard deviation σ_P allocates the fraction

$$w = \frac{\sigma_P}{\sigma_r}$$

of wealth to the risk asset and the remaining fraction $1 - w$ to the risk-free asset. Use this expression, together with your solution for the optimal choice of σ_P from question 3, above, to find the investors' optimal choice of w. How does the optimal share of the risky asset depend on the investor's risk aversion, as measured by the parameter A from the utility function? How does the optimal share of the risky asset depend on $\mu_r - r_f$, the risk premium defined as the expected return on the risky asset minus the risk-free return?